

JAXON COOPER

(707) 775-9028 | contact@jaxoncooper.com | linkedin.com/in/jaxoncooper | portfolio.jaxoncooper.com

EDUCATION

Cal Poly Humboldt

B.S. Mechanical Engineering, Upper Division Transfer

Aug. 2024 – Dec. 2026

Arcata, CA

- Awarded the *Outstanding Mechanical Engineering Graduate* scholarship.

EXPERIENCE

Undergraduate Research Assistant

Cal Poly Humboldt

Aug. 2025 – Present

Arcata, CA

- Prepare biological femur samples for experimentation via thawing, cutting to size, and labelling.
- Measure, mount, and mill femur samples on CNC.
- Execute and record drilling experiments.

Airport Ramp Technician

City of Petaluma

Aug. 2025 – Aug. 2026

Petaluma, CA

- Verified proper operation and quality of AVGAS and JET A fuel storage tanks, pumps, and fuel delivery trucks in compliance with safety procedures.
- Conducted runway inspections, aircraft fueling and marshalling, and airport grounds maintenance.

Auto Shop Assistant

Cal Poly Humboldt

Nov. 2024 – May 2025

Arcata, CA

- Supported maintenance of university fleet vehicles and buses through cleaning, refueling, and readiness checks.
- Maintained shop organization and inventory control.

Materials Engineering Lab Assistant

Santa Rosa Junior College

Dec. 2023 – May 2024

Santa Rosa, CA

- Assembled, calibrated, and organized materials testing equipment prior to the start of the semester.
- Troubleshooted and repaired faulty equipment throughout the semester.
- Guided students through testing equipment use during lab sessions.

PROJECTS

Capstone Project – High Precision Wood Wool Cutter

Aug. 2025 – May 2026

- Designed and manufactured a machine that shaves wood into precisely dimensioned wool shavings for use in sustainable building materials.
- Concentration on effective blade design, precise cutting and feed rate control, and meeting strand dimension constraints.
- Rigorous experimental design includes varying blade geometry and cutting parameters for best possible wood wool quality.

Rotorcraft Service Ceiling and Time-To-Climb Predictor Algorithm

Aug. 2025 – Dec. 2025

- Developed a Python program which makes service ceiling and TTC predictions based on rotorcraft specifications.
- Algorithm utilizes the Dormand-Prince ODE solving method to determine threshold altitude and TTC.

SKILLS

Technical Skills: GD&T, BOM Management, Metallography, HTML, CSS

CAD & Engineering Software: SOLIDWORKS, Autodesk Inventor, LabVIEW, LTspice

Manufacturing & Machining: CNC machining (G-code), Manual mill & lathe machining,

Testing & Measurement: Shimadzu UTM, 3-axis dynamometer, Thermal imaging, Oscilloscope, Waveform generator

Programming & Technical Computing: Wolfram Mathematica, LaTeX, Python, C, MATLAB, HTML, CSS

VOLUNTEERING

Boat Crewmember Trainee

U.S. Coast Guard Auxiliary

Apr. 2026 – Present

Petaluma, CA